

Evaluating the precision of eight spatial sampling schemes in estimating regional means of simulated yield for two crops

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ABSTRACT

We compared the precision of simple random sampling (SimRS) and seven types of stratified random sampling (StrRS) schemes in estimating regional mean of water-limited yields for two crops (winter wheat and silage maize) that were simulated by fourteen crop models. We found that the precision gains of StrRS varied considerably across stratification methods and crop models. Precision gains for compact geographical stratification were positive, stable and consistent across crop models. Stratification with soil water holding capacity had very high precision gains for twelve models, but resulted in negative gains for two models. Increasing the sample size monotonously decreased the sampling errors for all the sampling schemes. We conclude that compact geographical stratification can modestly but consistently improve the precision in estimating regional mean yields. Using the most influential environmental variable for stratification can notably improve the sampling precision, especially when the sensitivity behavior of a crop model is known.

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1. Introduction

Dynamic crop models are developed for simulating crop growth and yield in response to various environmental conditions and management practices at a field scale (Keating et al., 2003; van Diepen et al., 1989; Williams et al., 1989). To provide summarized information (e.g. mean/total crop production inside a political boundary) for agricultural impact and risk assessment to support policy making, crop models need to be applied over large areas. Due to data paucity and computing cost, simulations are typically conducted at a limited number of sample locations across a region, through which results are up-scaled to regional or larger scales (Ewert et al., 2011). For example, Rötter et al. (1995) chose 18 sites to represent a large watershed, the Rhine basin. Trnka et al. (2014) chose 14 sites to represent Europe to simulate the adverse weather events for wheat. Asseng et al. (2015) chose 30 sites across the world to simulate temperature effects on global wheat production. The methods used to select simulation locations, called sampling design, can be used to improve the representativeness of the simulation results (Roleček et al., 2007).

Many environmental characteristics show a spatial continuity, i.e. data at two nearby locations are on average more similar than data at two widely spaced locations. For this reason, when using environmental data as input to a crop model, the simulation results are spatially dependent (Caeiro et al., 2003). Despite this, classical sampling theory is perfectly valid for such spatially structured populations (Brus and De Gruijter, 1997; Brus and DeGruijter, 1993; De Gruijter and Ter Braak, 1990). Model-based and design-based are two widely used schemes of sampling (Cassel et al., 1977; Wang et al., 2013). For estimating global and regional means, design-based strategies can be advantageous (Brus and De Gruijter, 1993), while simple random sampling (SimRS) and stratified random sampling (StrRS) are two of the most important design-based strategies (Hirzel and Guisan, 2002; Ripley, 2005). In SimRS, a given number of sampling units are selected independently from each other and with equal inclusion probability (Cochran, 1977). In StrRS, the entire study area is separated into sub-regions, called strata (or zoning), frequently according to prior information on the population and then random sampling is applied to each stratum. These two design-based schemes have been widely evaluated in monitoring of natural resources (Brus, 1994; De Gruijter et al., 2006), species distribution modeling (Stockwell and Peterson, 2002; Wisz et al., 2008) and demographic health surveys (Kumar, 2007, 2009). In a vegetation survey, Austin and Heyligers (1989) found that stratifying the population by combined information on climate, topographic and lithological characteristics could better represent the environmental variability in the area, especially when the stratification is coupled with well-tuned sampling rules based on aspect and topographic position. Wang et al. (2002) found that zoning of the population based on prior knowledge of the influential variables could reduce the sample size to achieve the same efficiency in monitoring the area of cultivated land. Brus (1994) found that the estimation accuracy can be improved by stratifying the population based on soil and land use maps when estimating the spatial means of phosphate sorption characteristics. Wang et al. (2010) found that stratification of population in the study area could reduce the variance of estimators in surveys of non-cultivated land in China.

In these survey and monitoring applications, the prior information that is used to stratify the population is normally obtained from other correlated variables or historical survey data. In crop modeling, the output population is simulated with the input of environmental variables and management practices, which can be

used as prior information to create the strata. Many types of strata or zones including climate zones (Rötter et al., 2012), agro-climatic zones (Rötter et al., 1995), environmental zones (Metzger et al., 2005; Olesen et al., 2011), agro-ecological zones (Aggarwal, 1993), and climate-soil zones (Bryan et al., 2014; Zhao et al., 2015a), have been used for regional or global crop modelling studies. However, only very few studies explicitly investigated the precision of these stratification methods and spatial sampling strategies. Nendel et al. (2013) showed that one soil profile and weather station were not sufficient to represent the observed mean grain yields of winter wheat in Thuringia, a region in Germany covering more than 16 000 km². By using one soil profile and gridded weather data at 1 km spatial resolution, van Bussel et al. (2016) evaluated the effects of sample size of StrRS on simulations of winter wheat yields under two production conditions, i.e. potential and water-limited in North Rhine-Westphalia. They recommended that detailed soil properties should be included in the simulations to further consolidate the conclusions from their study. To our best knowledge, no study has compared the efficiency of different stratum types (i.e. variables used to create the strata) and stratum number for estimating regional mean of simulated crop yields.

This study aims to compare the precision of SimRS and seven types of StrRS in estimating regional mean yield for two crops (winter wheat and silage maize). We investigated how the precision, indicated by mean squared error (MSE), depends on the sample sizes, the variables used for stratification, the number of strata, the crop types and the crop models.

2. Methods

2.1. Sampling precision

Crop yields of a region (A) constitute a continuous surface that can be infinitely divided. However, due to computing cost and input data availability, it was not possible even to do the simulations for each individual field of the entire study area. Instead, we divided the A into 1×1 km grid cells and simulated yield for each cell. The results were treated as the full population ($N = 34,168$) and the average over all cells was treated as the true regional yield $\bar{Y}(A)$. We sampled the population with a range of sample sizes and sampling schemes to giving various estimates $\hat{\bar{Y}}(A)$ of $\bar{Y}(A)$.

Eight design-based sampling schemes were evaluated, including simple random sampling (SimRS) and seven stratified random sampling (StrRS) with strata based on different environmental variables. A stratification method with L strata divides the population of grid cells into L non-overlapping groups. SimRS can be treated as a one stratum StrRS ($L = 1$). For any particular stratification method, let N_h denote the number of cells within stratum h . This is determined by the stratification scheme, and is known. Suppose from each stratum a simple random sample without replacement is selected. The sample size within stratum h is noted n_h . The symbols used in this study are shown in Table 1.

The estimated mean using stratified random sampling $\hat{\bar{Y}}(A)$ was calculated as

$$\hat{\bar{Y}}(A) = \frac{\sum_{h=1}^L N_h \hat{\bar{Y}}_h}{N} = \sum_{h=1}^L w_h \hat{\bar{Y}}_h \quad (1)$$

where $\hat{\bar{Y}}_h$ is average yield in stratum h , estimated using the samples from that stratum.

The estimator $\hat{\bar{Y}}_h$ is unbiased, since the mean of all possible samples equals to the true population mean of stratum h . Therefore, $\hat{\bar{Y}}(A)$ is also an unbiased estimator of the population mean $\bar{Y}(A)$ of the entire region according to Theorem 5.1 in Cochran (1977). To

Table 1

Symbols are used for simple random sampling and stratified random sampling in this study.

Symbol	Description	Symbol	Description
N	Population size, total number of grid cells, in this study $N = 34,168$	y_{hi}	Yield for the i th sample in stratum h
n	Sample size for region A	$w_h = \frac{N_h}{N}$	Weight of stratum h
$\bar{Y}(A)$	True population mean for region A	$f_h = \frac{n_h}{N_h}$	Sampling fraction in stratum h
$\hat{\bar{Y}}(A)$	Population mean estimator	$\bar{Y}_h = \frac{\sum_{i=1}^{n_h} y_{hi}}{N_h}$	True mean yield of stratum h
L	Numbers of strata, in this study $L = 4, 8$ or 16	$\hat{\bar{Y}}_h = \frac{\sum_{i=1}^{n_h} y_{hi}}{n_h}$	Mean yield estimator in stratum h
N_h	Number of grid cells in stratum h	$S_h^2 = \frac{\sum_{i=1}^{n_h} (y_{hi} - \bar{Y}_h)^2}{N_h}$	True variance in stratum h
n_h	Sample size in stratum h		

quantify the sampling error of a sampling scheme we used the mean squared error

$$MSE = E \left[\left(\hat{\bar{Y}}(A) - \bar{Y}(A) \right)^2 \right] \quad (2)$$

Since $\hat{\bar{Y}}(A)$ is unbiased, $E[\hat{\bar{Y}}(A)] = \bar{Y}(A)$. The mean squared error equals the variance (Wackerly et al., 2007)

$$\begin{aligned} MSE &= Var \left[\hat{\bar{Y}}(A) - \bar{Y}(A) \right] + Bias_{\bar{Y}(A)}^2 = Var \left(\hat{\bar{Y}}(A) \right) \\ &= Var \left[\sum_{h=1}^L w_h \hat{\bar{Y}}_h \right] \end{aligned} \quad (3)$$

Since the samples in different strata are drawn independently, the covariance between strata equals 0. Therefore, MSE can be calculated as

$$\begin{aligned} MSE &= \sum_{h=1}^L \left[w_h^2 Var \left(\hat{\bar{Y}}_h \right) \right] + 2 \sum_{k=1}^L \sum_{j>k}^L w_h w_j Cov \left(\bar{y}_h \bar{y}_j \right) \\ &= \sum_{h=1}^L \left[w_h^2 Var \left(\hat{\bar{Y}}_h \right) \right] \end{aligned} \quad (4)$$

According to Theorem 2.2 in Cochran (1977), $Var(\hat{\bar{Y}}_h) = \frac{S_h^2}{n_h} \frac{N_h - n_h}{N_h}$, and so Eq. (4) gives

$$\begin{aligned} MSE &= \frac{1}{N^2} \sum_{h=1}^L N_h^2 Var \left(\hat{\bar{Y}}_h \right) = \frac{1}{N^2} \sum_{h=1}^L N_h (N_h - n_h) \frac{S_h^2}{n_h} \\ &= \sum_{h=1}^L w_h^2 (1 - f_h) \frac{S_h^2}{n_h} \end{aligned} \quad (5)$$

To facilitate the interpretation of the results, we used the root mean squared error, $RMSE = \sqrt{MSE}$, to measure the sampling precision of a sampling scheme. The true variance, weight and sample size in each stratum were input into Eq. (5) to derive the MSE , and later $RMSE$ for each sampling scheme and sample size.

Finally, we calculated the precision gain (PG) of different types of stratification against $SimRS$ as

$$PG = \frac{RMSE_{SimRS} - RMSE_{StrRS}}{RMSE_{SimRS}} \times 100 (\%) \quad (6)$$

To ensure a fair comparison, $RMSE_{SimRS}$ and $RMSE_{StrRS}$ always had the same sample size.

2.2. Study area

The study area, the state of North Rhine-Westphalia (NRW, 6 E – 9.5 E, 50 N–52.5 N), is located in the middle-west of Germany (Fig. 1). The study area locates in a temperate and humid climate

zone of Central Europe with maritime influence from the Atlantic Ocean. Flat plain covers 50% of the study area. The topography rises from northwest towards the southeast of the state and merges into Germany's Central Uplands. Agriculture land occupies more than 60% of the state area. Winter wheat and silage maize are the pre-dominant crops according to the yield report of the Federal Statistical Office (2013). We simulated crop growth and yields over the entire region (34,168 grid cells) without considering actual land use in reality. Thus the simulations represented a region with large heterogeneity of environmental conditions but including some areas which may not be suitable for crop growth.

2.3. Climate and soil

Thirty years (1982–2011) gridded monthly weather data including maximum, mean, and minimum temperature, sunshine hours, and daily precipitation, at 1 km resolution were obtained from the German Meteorological Service (DWD, 2015). The data were combined with station-based daily data from more than 200 local weather stations to produce the gridded daily weather data as model inputs. The detailed procedures for fusion of the two data sources were described in Zhao et al. (2015b) and Siebert and Ewert

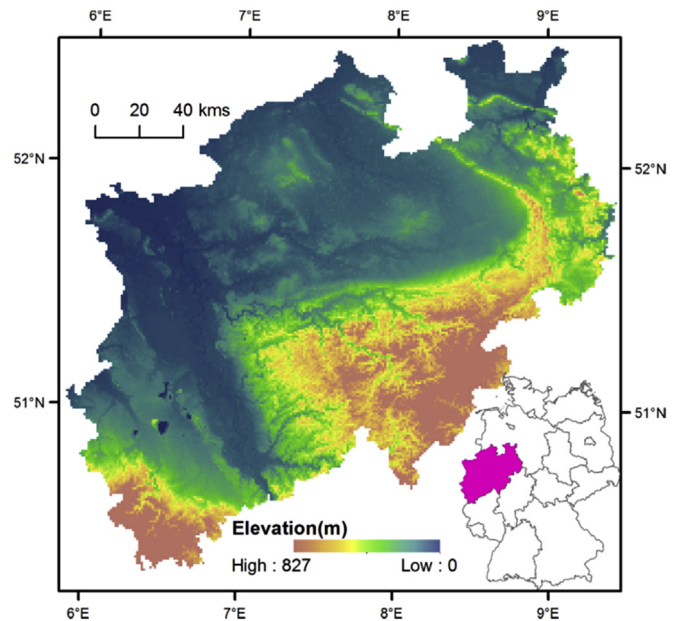


Fig. 1. The location and terrain of the study area, North Rhine-Westphalia (NRW), Germany (data source: Federal Agency for Cartography and Geodesy, Germany, <http://www.bkg.bund.de>). The colored region in the map of States of Germany indicates the location of NRW. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(2012). A summary of daily mean temperature, annual sum precipitation and annual sum global solar radiation can be found in Fig. 2.

The soil data at a scale of 1: 50,000 were obtained from GDNRW (2001). The soil data available in vector format were converted to a raster format of 300 m spatial resolution. To convert the soil data from 300 m to 1 km spatial resolution, we took the area-dominant soil types of the 300 m grid cells inside the 1 km $\times$ 1 km grid cell and allocated the profile of the dominant soil type to the 1 km resolution grid cells (Fig. 2e). The profiles of different physical properties sharing the same soil type were considered as unique soil types in the aggregation. The soil water holding capacity

(SWHC) and the dominant soil types for each 1 km $\times$ 1 km grid cell are shown in Fig. 2(d and e).

2.4. Crop models and simulations set up

Fourteen process-based crop models participated in this study: AGROC (Bauer et al., 2012), APSIM (Holzworth et al., 2014; Keating et al., 2003), APSIM-NWHEAT (Asseng et al., 2000), CENTURY (Kelly et al., 1997; Parton and Rasmussen, 1994), CropSyst (Stöckle et al., 2003; Stöckle et al., 1994), CoupModel (Conrad and Fohrer, 2009; Jansson, 2012), DailyDayCent (Del Grosso et al., 2006; Yeluripati et al., 2009), EPIC (Williams et al., 1983; Williams and Singh,

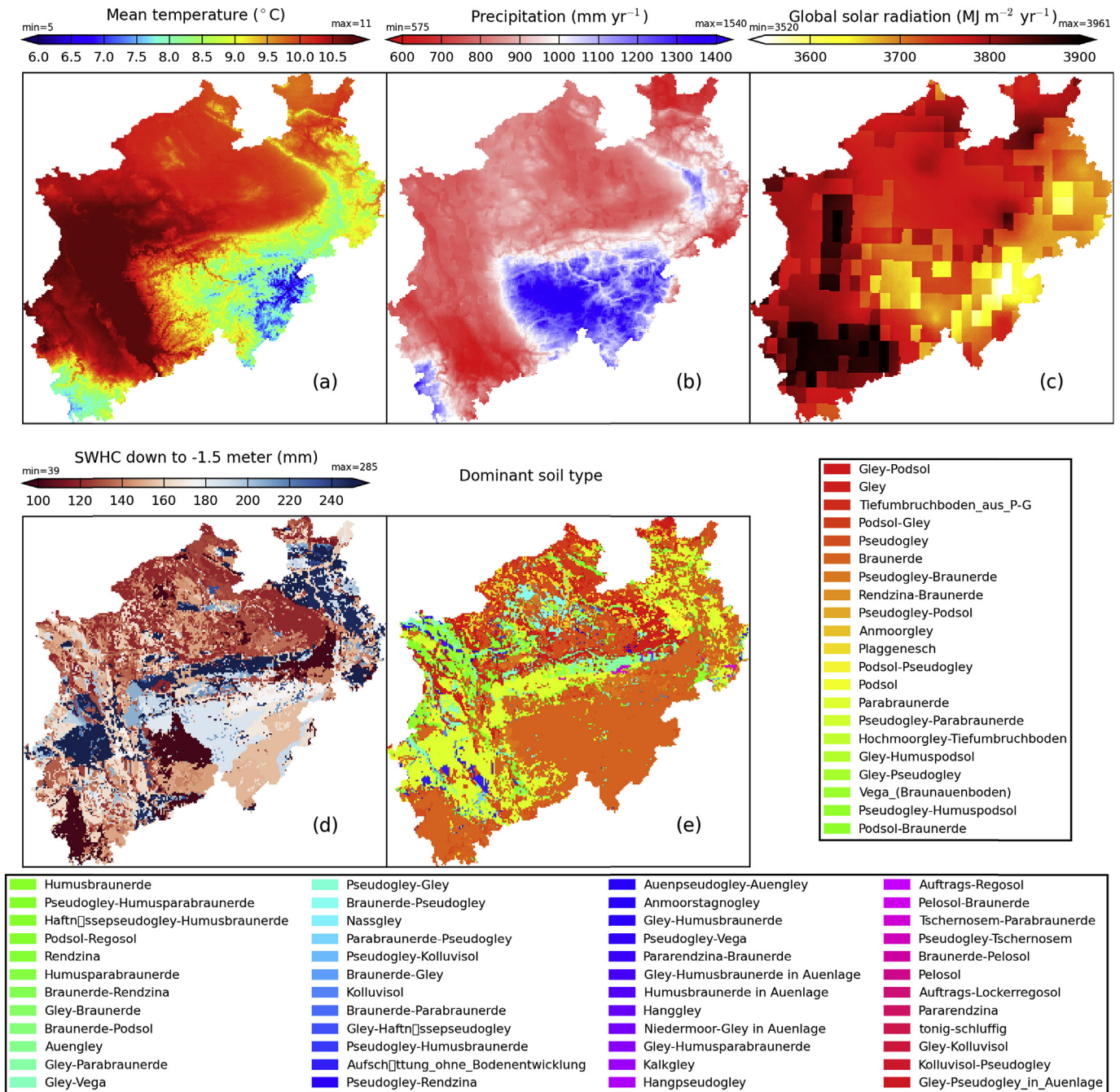


Fig. 2. Summary of the climate and soil data at 1 km spatial resolution for the study area, North Rhine-Westphalia (NRW), Germany. The climate variables include average of daily mean temperature (a), annual sum precipitation (b) and annual sum radiation (c). The soil property variables include soil water holding capacity (SWHC) down to -1.5 m depth (d) and area-dominant soil types at 1 km spatial resolution (e).

1995), Expert-N (Klein et al., Submitted; Priesack et al., 2007), HERMES (Kersebaum, 2007), SIMPLACE<LINTUL> (Gaiser et al., 2013; Zhao et al., 2015b), MCWLA (Tao et al., 2009; Tao and Zhang, 2013), MONICA (Nendel et al., 2011) and STICS (Bergez et al., 2014; Brisson et al., 2003). All 14 models were used to simulate the yields of winter wheat (Fig. 4) and 10 models simulated silage maize (ten models, Fig. 5) across NRW (34,168 grid cells). Table S1 in the supplementary gives details about the models.

The simulated cropping systems were continuous winter wheat or continuous silage maize over the entire area. Crop production could be limited by water availability. For winter wheat, 400 plants m^{-2} were sown at the depth of 4 cm on 1st October. For silage maize, 10 plants m^{-2} were sown at the depth of 6 cm on 20th April. Typical harvest date for winter wheat (1st Aug) and silage maize (20th Sep) was provided to calibrate the phenology. The historical mean yields (1999–2011) of NRW (Federal Statistical Office, 2013) for the two crops were used to calibrate the biomass production.

2.5. Stratification and precision gain quantification

Stratification was conducted based on coordinates (compact geographical stratification (Brus et al., 1999)), temperature, precipitation, radiation, climate (temperature, precipitation and radiation), soil (SWHC down to -1.5 m depth) or environmental conditions (climate, soil and terrain) (Fig. 3). In each case, the stratification variable(s) were fed into a *k-means++* clustering algorithm and the clustering results were used as strata (Arthur and Vassilvitskii, 2007). The *k-mean++* algorithm was used for initialization of the clustering (Pedregosa et al., 2011). For each clustering, it was run with 100 different centroid seeds and the maximum number of iterations for a single run was set as 1000. Since the variables that were used for stratification had different orders of magnitude, they were first normalized to zero mean and unit variance. For each type of stratification, $L = 4, 8$, or 16 strata were created. In the SimRS, the entire study area was treated as one

stratum ($L = 1$) and the samples were randomly drawn from the population of simulated yields. In the seven StrRS schemes, the sample size was identical in each stratum, $n_h = n/L$. We tested sample sizes from 2 per stratum up to a total sample size of 200. For example, for $L = 8$, the tested total sample sizes were $L*2 = 16$, $L*3 = 24$, ..., $L*25 = 200$.

3. Results

3.1. Simulated yields by different models and the models ensemble mean

For winter wheat, the simulated yields (population) by APSIM, APSIM-NWHEAT, EPIC, Expert-N, SIMPLACE<LINTUL> and STICS were higher than by CropSyst and HERMES (Fig. 4q), which underestimated yields for a larger fraction of the region (Fig. 4f and i). The majority of the models simulated yields with very high spatial variability. The spatial patterns of simulated crop yields were consistent for the majority of the models. The yields simulated by APSIM, CropSyst, DailyDayCent, EPIC, Expert-N, HERMES, SIMPLACE<LINTUL> and STICS were strongly affected by the soil and showed extremely low yields across the regions with low water holding capacity (SWHC) (Figs. 2 and 4). The yields simulated by AGROC, APSIM-NWHEAT, CENTURY, CoupModel, MCWLA, and MONICA were less sensitive to SWHC compared to other models (Fig. 4a, c–e, j and k).

For silage maize, the simulated yields by APSIM, CENTURY, CropSyst, HERMES, MONICA, SIMPLACE<LINTUL> and STICS were higher than by AGROC and EPIC, which showed a considerable underestimation (Fig. 5). There were two contrasting spatial patterns across the models. For example, APSIM, CropSyst, MONICA, SIMPLACE<LINTUL> and STICS simulated low yields in the south-eastern mountainous regions, while CENTURY, DailyDayCent and HERMES simulated relatively high yields. The yields simulated by the models APSIM, AGROC, EPIC, HERMES, SIMPLACE<LINTUL> and

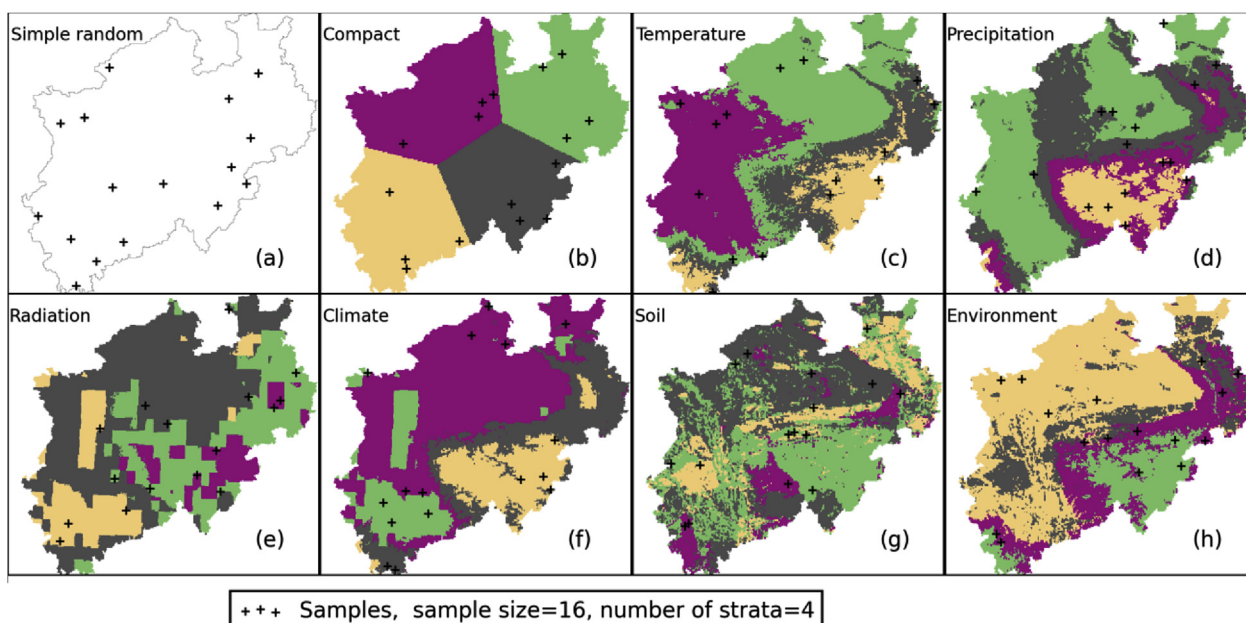


Fig. 3. Illustration of eight sampling schemes: simple random (a), stratification of coordinates, compact geographical stratification (b), stratification of temperature (c), stratification of annual precipitation (d), stratification of annual global radiation (e), stratification of climate conditions (f), stratification of soil (g) and stratification of environmental conditions (climate, soil and terrain) (h). In the stratified random sampling, four different strata are indicated by different colors. The stratifications were created by *k-means++* clustering of the corresponding variables. The stratifications according to climate conditions are clustered according to a combination of three climate variables (temperature, precipitation and radiation). The soil stratifications are clustered according to soil water holding capacity down to -1.5 m. The stratifications of environmental conditions are clustered according to a combination of climate, soil and elevation. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

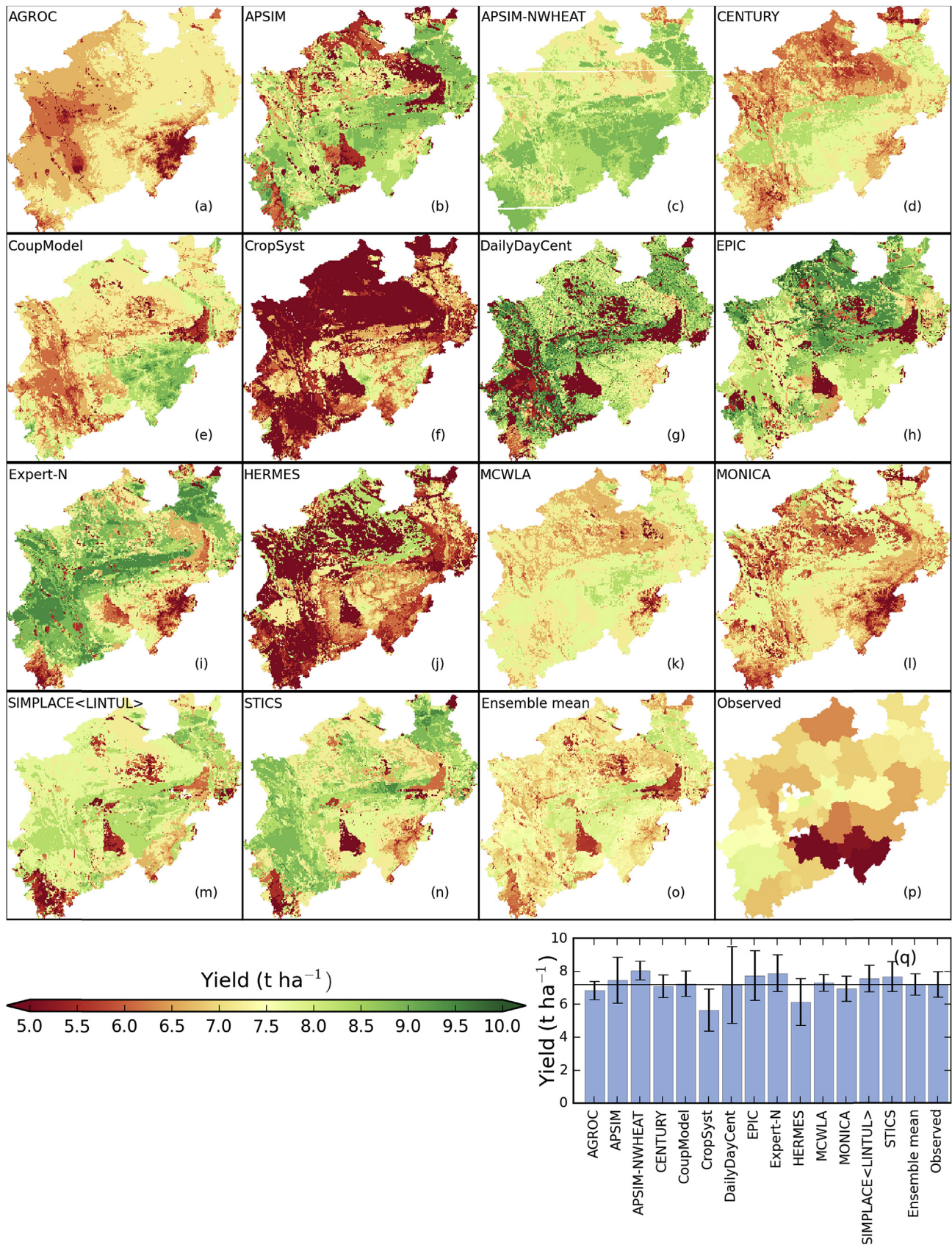


Fig. 4. Spatial distributions of simulated winter wheat yields by fourteen crop models (a–n), the models ensemble mean (o), mean observed yields (1999–2011, 7% of moisture in the reported values) (p), and the mean and standard deviation of wheat yields across grid cells (q). The simulations were conducted under water limited conditions from 1982 to 2011. The observed yields were reported at district levels from 1999 to 2011. In the bar plot (q), the bar height indicates the mean yield across all grid cells and the error bar indicates the standard deviation. The horizontal line indicates the mean observed yield over the study area.

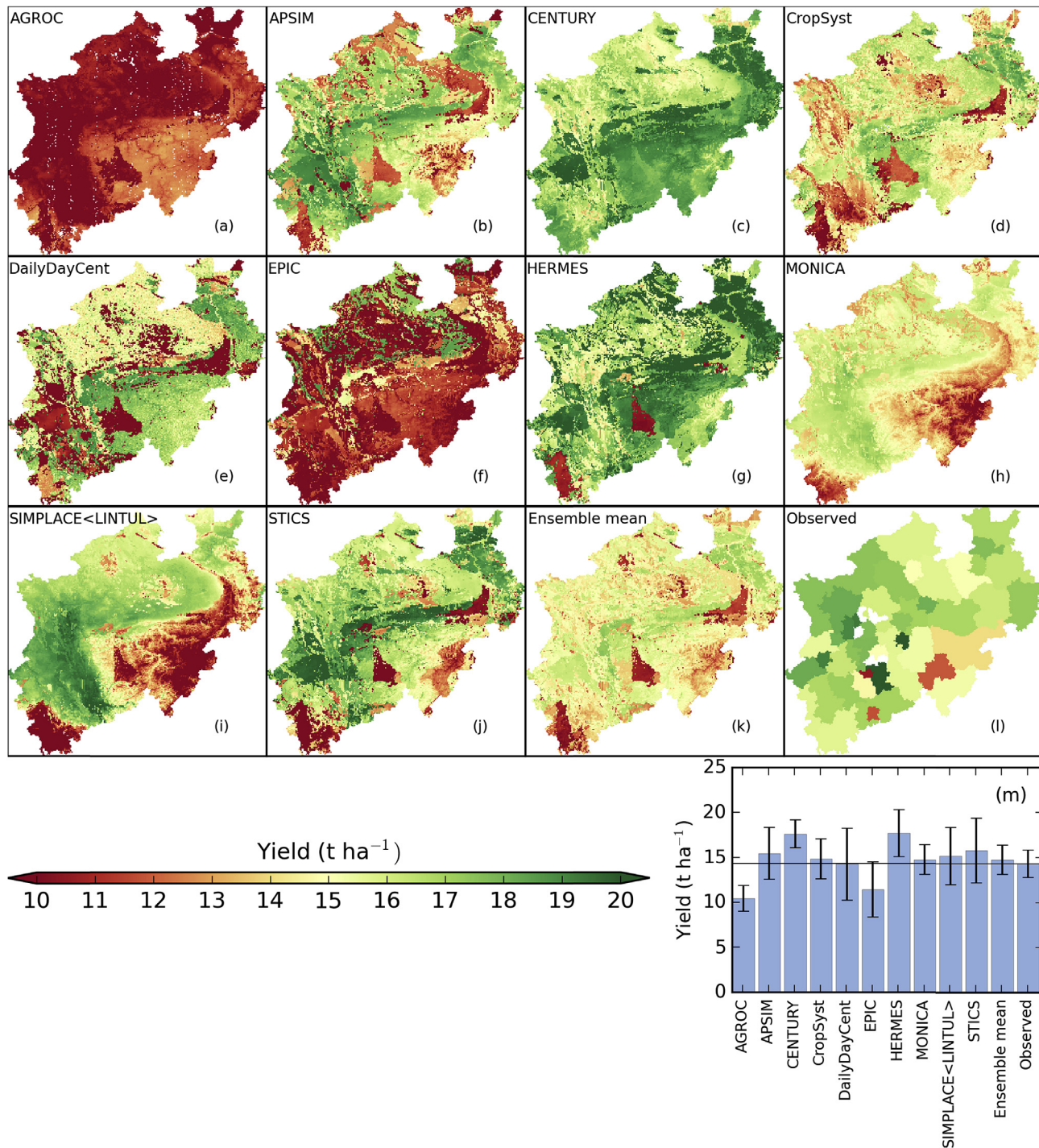


Fig. 5. Spatial distributions of simulated silage maize yields by ten crop models, models ensemble mean, and mean observed yields (1999–2011, 7% of moisture in the reported values) (a–i), and the mean and standard deviation of silage maize yields across grid cells (m). The simulations were conducted under water limited conditions from 1982 to 2011. The observed yields were reported at district levels from 1999 to 2011. In the bar plot (m), the bar height indicates the mean yield across all 34,168 grid cells and the error bar indicates the standard deviation. The horizontal line indicates the mean observed yield over the study area.

STICS strongly responded to low SWHC (Figs. 2 and 5), while yields simulated by CENTURY and MONICA showed a relatively low sensitivity (Fig. 5c and h).

3.2. Sampling precision across sample sizes

In estimating the regional mean yield $\bar{Y}(A)$ for winter wheat with SimRS, sampling standard errors decreased monotonously

from 0.750 to 0.075 t ha⁻¹ with the increase of sample size (Fig. 6b). With StrRS and $L = 16$ strata, the sampling standard errors decreased with similar slopes for all the seven stratification methods (Fig. 6a). The ranking of the stratification methods was the same for all sample sizes. Stratification based on soil water holding capacity had the smallest errors and has the highest precision. Stratification based on environmental conditions ranked the second and compact geographical stratification

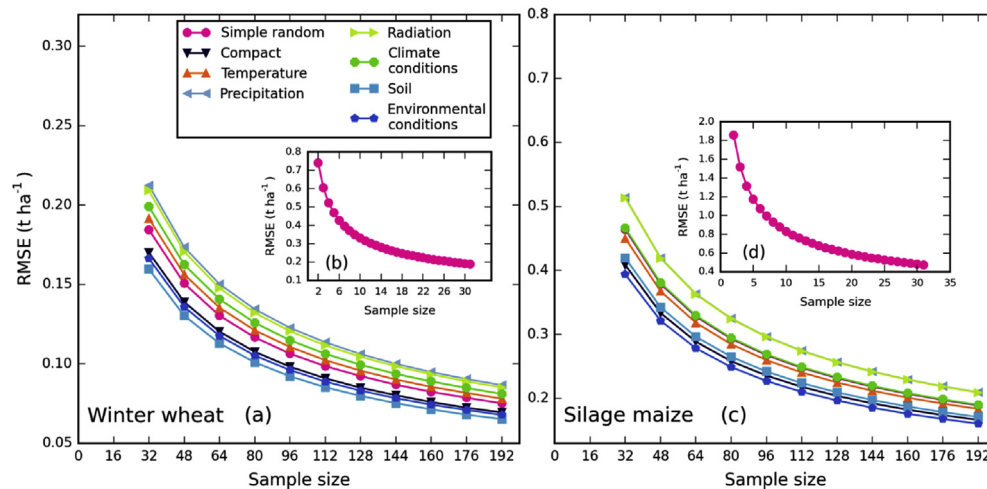


Fig. 6. The variation of sampling standard error ($RMSE$) of eight spatial sampling schemes with the increase of sample sizes (2–200 for simple random sampling (SimRS) and 32 to 200 for stratified random sampling (StrRS)) for the mean yield of two crops, winter wheat (a) and silage maize (c). The sub-plots (c) and (d) inside (a) and (b) show data for SimRS from 2 to 32. The sampling standard error of a sampling scheme is indicated by the root mean squared error ($RMSE$). The stratum number for the StrRS is 16. To use Eq. (5) to calculate the sampling standard error, each stratum needs at least two samples so that the smallest sample size for the StrRS is 16.

ranked the third. Stratification based on precipitation had the largest errors and lowest precision.

In estimating the regional mean yield $\bar{Y}(A)$ for silage maize with SimRS, sampling standard errors decreased monotonously from 1.90 to 0.19 $t\ ha^{-1}$ with the increase of sample size (Fig. 6d). With StrRS and $L = 16$ strata, the sampling standard errors decreased with similar slopes for all the seven stratification methods (Fig. 6c). As for wheat, the ranking of the stratification methods was the same for all sample sizes. Stratification based on environmental conditions always had the smallest standard errors and highest precision. Stratification based on soil and compact geography had the same sampling standard errors and the precision for them ranked second. Stratification based on radiation and precipitation had the largest sampling standard errors and least precision.

3.3. The precision gain of the stratified random sampling

In estimating the regional mean yields $\bar{Y}(A)$ for both crops, StrRS with compact geographical stratification always had positive precision gains (PG) and the PG increased when increasing the number of strata from 4 to 16 (Fig. 7a and b). With 16 strata, the compact geographical stratification can gain precision by 8% (median) for winter wheat and 10% (median) for silage maize. Stratification with temperature, precipitation, radiation and climate conditions did not improve the sampling precision, but rather had negative effects. For both crops, the medians of PG for these stratifications were negative, varying from -16% to -2% . The stratification based on environmental conditions with 16 strata achieved the highest PG . The improvement in sampling precision was 15% (median) for winter wheat and 26% (median) for silage maize. The stratification based on environmental conditions with 8 strata achieved the second highest PG . The medians of improvements in sampling precision were 15% (median) for winter wheat and 10% for silage maize. The stratification based on SWHC with 8 strata achieved the third highest PG . The medians of improvements in sampling precision were 11% (median) for winter wheat and 8% for silage maize. The increase of number of strata improved the PG for stratifications based on compact geography, soil and environmental conditions.

3.4. Comparing precision gains of the stratified random sampling across crop models

The precision gains (PG) of StrRS with compact geographical stratification were positive across all the fourteen crop models (Fig. 8). For compact geographical stratification, the largest PG for winter wheat was obtained with APSIM-NWHEAT (19%) and for silage maize with SIMPLACE<LINTUL> (32%) (Fig. 8a). For winter wheat, stratifying the study area based on temperature led to negative gains for the majority of the crop models, except for AGROC (53%), CoupModel (14%), and Expert-N (6%). For silage maize, it led to very high gains for SIMPLACE<LINTUL> (56%) and MONICA (40%), but not for other models. Similarly, stratifications with precipitation, radiation and climate conditions led to negative gains for the majority of models. However, stratification with soil led to high positive gains for most of the models for both crops. Stratifications with environmental conditions resulted in relatively large negative gains for MONICA (winter wheat) and EPIC (silage maize).

4. Discussion

Despite the frequent use of crop models for regional/global change assessments, the access to weather, soil profile, crop management and yield observations is often limited. Since the simulation quality is always affected by the amount and quality of observed data for model input, calibration and validation, elaborating the sampling scheme in order to select the most representative sites is essential for improving the efficiency and precision of crop modeling over large areas (Wisz et al., 2008). With an efficient sampling scheme, efforts on experiment implementation and data gathering can target at the most representative sites, thus the cost can be reduced. Although stratification of study area is widely adopted for regional crop modeling, very few previous studies have explicitly tried to quantify the efficiency of and optimize the sampling design. In this study, we tested the precision of eight spatial sampling schemes in estimating the regional mean yields $\bar{Y}(A)$ for two crops. The methods used to create the strata in stratified random sampling (StrRS) and the derived knowledge referring to the precisions of different sampling schemes is useful for up-scaling of crop models using a limited number of representative

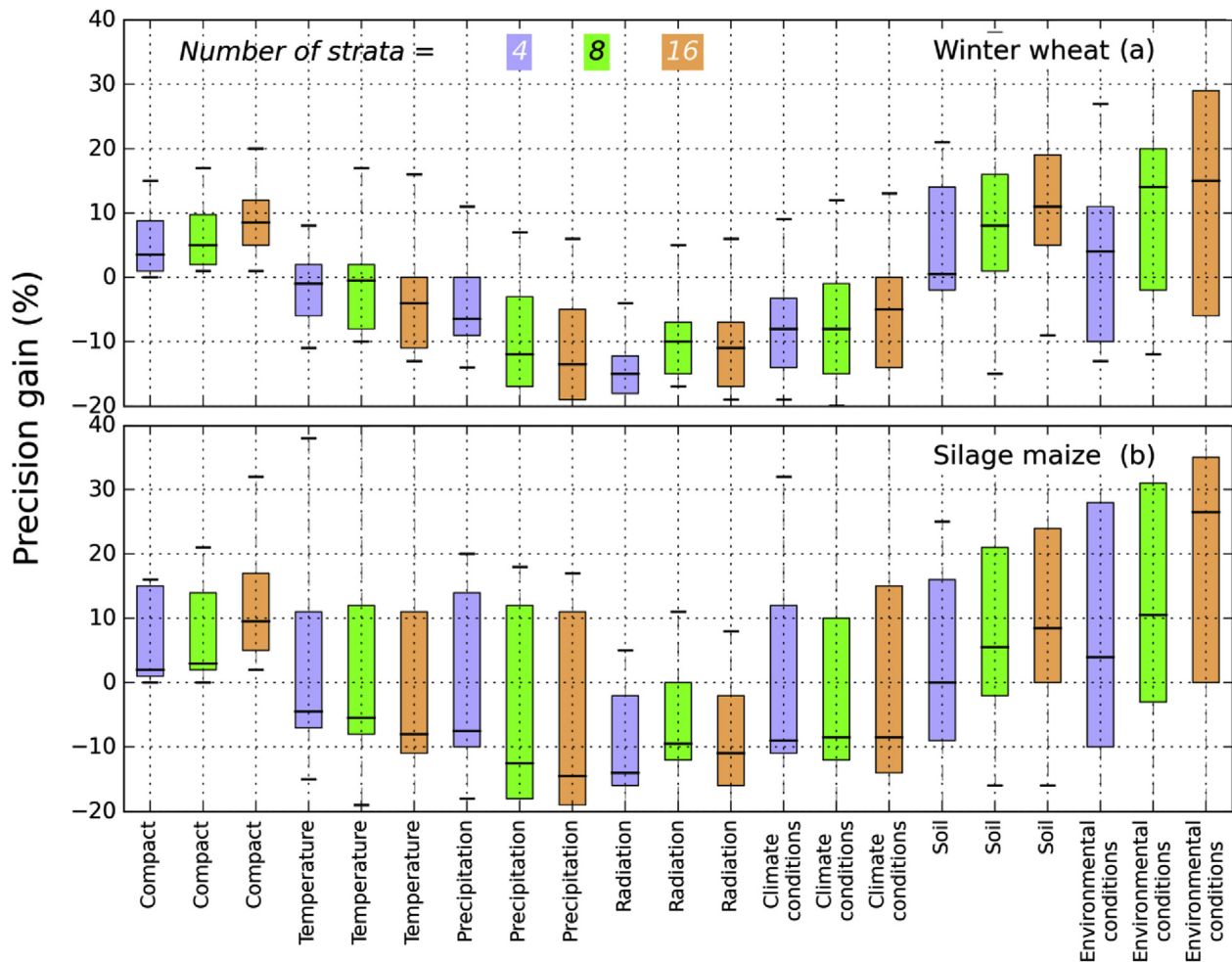


Fig. 7. The precision gains of stratified random sampling for two crops, winter wheat (a) and silage maize (b). The numbers of strata for the different stratified random sampling schemes are 4, 8 and 16, which are indicated by three different colors as shown in the legend. The variations in precision gain result from different crop models (fourteen for winter wheat and ten for silage maize) and sample sizes (32, 48, ..., 192). The edges of the box are the lower hinge (the 25th percentile, Q1) and the upper hinge (the 75th percentile, Q3), and the whiskers extend to $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$, $IQR = 1.5 \times (Q3 - Q1)$. The horizontal black line in the box represents the median of precision gains. The precision gain is calculated with Eq. (6). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

locations. The results from this study can guide the choice of simulation sites when up-scaling crop models by conducting simulations at a limited number of sites.

4.1. Simple random sampling (SimRS) versus stratified random sampling (StrRS)

The simple random sampling (SimRS) performs best when population is relatively homogeneous and evenly distributed. This is rarely the case for environmental variables across wide geographical areas where spatial heterogeneity and autocorrelation is ubiquitous (Overmars et al., 2003). For example, there is a large block of land with soils of very low water holding capacity in the southeast of our study area (Fig. 2). Therefore, the SimRS needs a large sample size to attain the same precision as StrRS (Fig. 6) (Fortin et al., 1989). This is particularly true when the environmental gradients are distributed non-randomly in space and similar samples are clumped together. In line with this study, Mohler (1983) showed that SimRS could result in truncated response curves for sampling species distribution especially when the extremes of the major environmental gradients were missed in the sampling. Stratifying along these gradients and being particularly careful about sampling the extremes can guarantee an

efficient sampling of these outer limits.

To avoid the drawbacks of SimRS, StrRS divides the population into relative homogeneous subgroups so that a smaller sample size is required for the same precision. It fulfills the requirement that efforts are not wasted for simulating yields across similar environmental conditions for applications of regional crop modeling (Danz et al., 2005). According to Eq. (5), the sampling standard error (RMSE) is determined by the population size, sample size and the true variance in each stratum. Increasing the sample size can continually decrease the sampling standard error but requires more resources and becomes more expensive in gathering the input data and calibration of the models. The computing cost of a large number of simulations can be overcome by taking advantage of advanced hardware and parallel computing (Zhao et al., 2013), but long-term records of weather data and soil profile measurements are not available for many locations. Therefore, increasing the sample size is not always suitable to improve the precision of regional crop model applications. Another possible way of increasing the estimation precision is reducing the population variance (S_h^2) in the strata. The smaller the variance in each stratum, the more precision is gained by a StrRS sampling scheme. However, the variances cannot always be reduced by stratification (Cochran, 1977). This aspect was demonstrated by the results from this study,

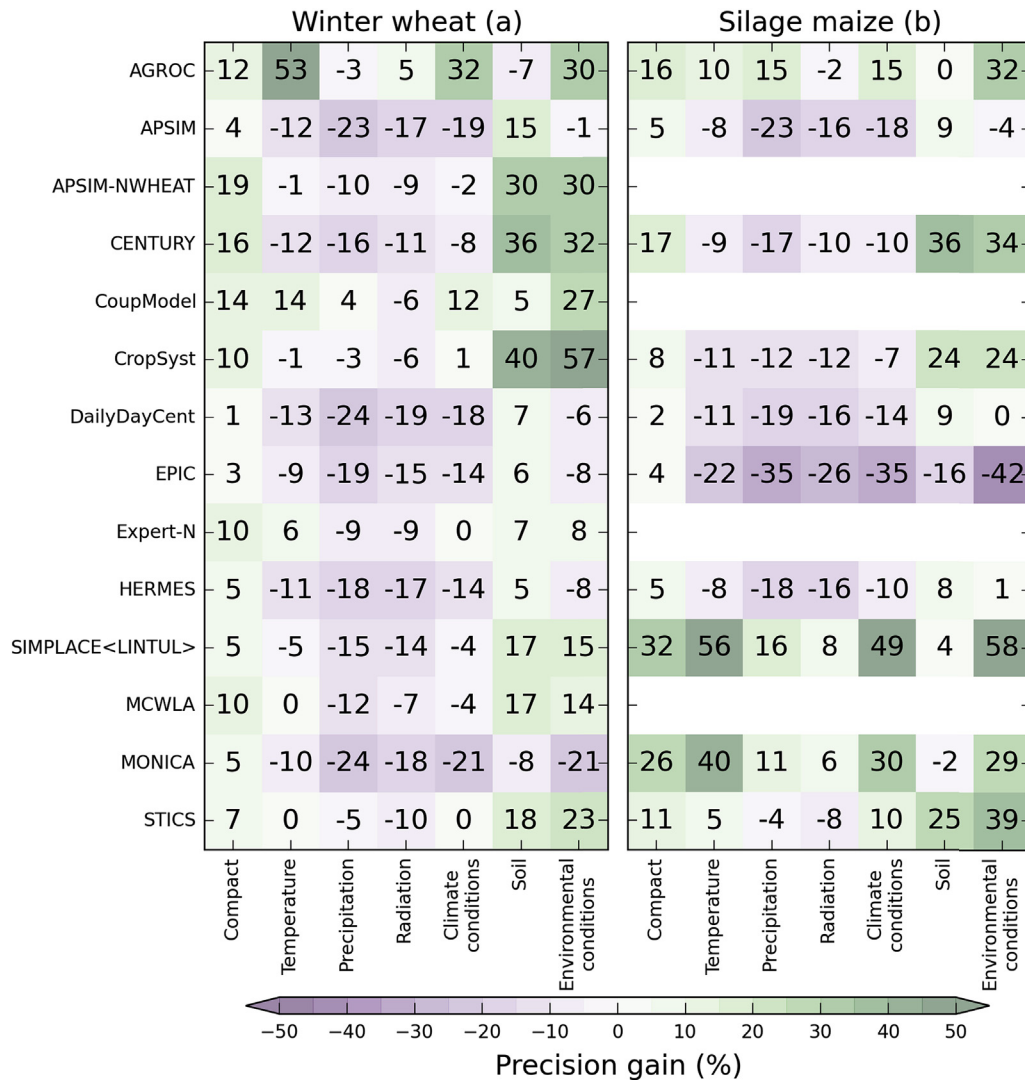


Fig. 8. The precision gain (%) of seven stratified random sampling of fourteen crop models in estimating regional mean yields of two crops, winter wheat (a, left) and silage maize (b, right). The number of strata for the different stratified random sampling schemes is 16. The sample sizes include 32, 48, ..., 192. The precision gain values in the plots are the mean for a crop model in combination with a stratification type across different sample sizes. The precision gain is calculated with Eq. (6).

which showed that many stratification types resulted in negative gains (Figs. 7 and 8). The reason for the negative gains was that inappropriate prior information was used for the stratification of the population, which did not result in a homogeneous population within the strata. When evenly assigning the samples to strata, many samples were wasted in the relative homogeneous strata and, at the same time, the sample size was too small for many heterogeneous strata. A optimized allocation of the samples may further improve the precision of the inefficient stratification (Cochran, 1977).

For a specific combination of stratum number and sample size, the environmental variables used to create the strata determine the stratum shape, and thus the population variance in the strata. If the stratification could reduce the heterogeneity in each stratum, the variance of sampled yields in each stratum can be minimized (Caeiro et al., 2003). Therefore, StrRS has a promising potential to improve the regional crop modeling when spatial autocorrelation of the simulated yields is obvious and strong. The results from this study showed that the choice of prior environmental information is critical for the sampling precision of the StrRS. The stratification based on environmental

conditions reduced the sampling error for both of the simulated crops for most of the models (Figs. 7 and 8). This is due to the fact that the yields for the two crops were simulated under water-limited conditions and many of the crop models are sensitive to SWHC. The results imply that only the most influential environmental variable(s) should be used to create the strata in order to minimize the sampling error of StrRS, which is also in agreement with the conclusion from Wang et al. (2002). Since the sensitivity to different input variables varies across crop models, the efficiency of using a certain type of strata based on prior information of the same variable might differ across crop models.

4.2. Variation of precision gains across crop models

In estimating the mean yields for the two crops, the precision gain of the same StrRS varied across models. This may be due to the different response sensitivities of models to the environmental variables that were used to create the strata (Fig. 8). If the model is not sensitive to the variable(s) that is (are) used to create the strata, the stratification cannot reduce the heterogeneity in the strata. In this case, stratification cannot improve precision, and may even

cause negative gains. To verify that the variation of the performance was caused by model sensitivity differences, we conducted a sensitivity analysis of simulated yields to the environmental variables that were used to create the strata (Fig. 9). The results verified our assumption. For example, in estimating the mean yields for maize, SIMPLACE<LINTUL> had a high precision gain with the stratification of temperature (56%), precipitation (16%) and radiation (8%), but only small gain with soil (4%). Correspondingly, the simulated yields of silage maize with SIMPLACE<LINTUL> had a strong correlation with temperature ($|\rho| = 0.9$), precipitation ($|\rho| = 0.63$) and radiation ($|\rho| = 0.59$), but only a weak correlation with soil ($|\rho| = 0.21$). Similar agreement between the variables most useful for stratification and the variables to which a model was most sensitive was also found for other models. Therefore, the strata shape should be based on those environmental variables to which the model output of interest is most sensitive.

In the present study, the mean of the simulated yields across all the 1 km grid cells were the population true mean ($\bar{Y}(A)$). Due to the differences in the models' structure, the environmental variables that should be considered for selection of the simulation sites are crop model specific. Model sensitivity can be analyzed only when the simulations are executed. At the same time, the strata need to be created according to the model sensitivity behaviors. This issue can be solved by a two-step method which has been applied to the natural resource survey (Guisan et al., 2006). First, a

crop model can be executed across some random locations/sites to study the model sensitivity. Then, the study area can be stratified according to the one of the most sensitive variables derived from the previous steps.

4.3. Limitations

Crop yields were simulated over the entire region without considering actual land use. However, the lack of a realistic land use pattern is compensated by the advantage of testing the methods over a wide range of conditions. For large scale studies (e.g. global scale), the crops can be cultivated in severe conditions (e.g. in middle Asia and Africa). Thus, allowing a larger heterogeneity of environmental conditions here might be more informative about the best sampling strategy for large scale studies.

This study focused on the sampling precision for yields simulated under water-limited conditions. In reality, the yields are also influenced by a number of additional abiotic and biotic factors and by management practices. The interactions between environmental variables and management practices could change the sensitivity of the simulated yields to different environmental variables (Zhao et al., 2014). Considering the stress related to nitrogen availability could enhance the yield's sensitivity to the initial soil organic matter and nitrogen.

In StrRS, proportional and optimal allocations of the samples

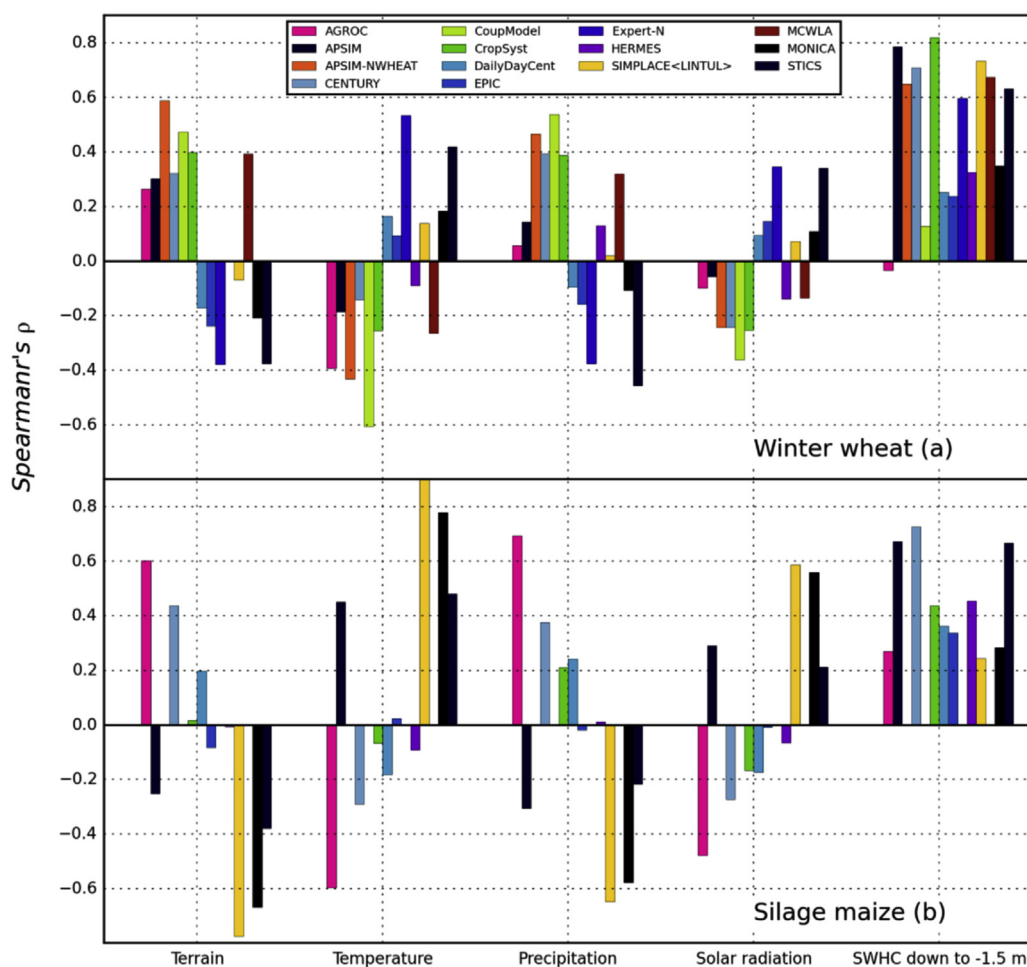


Fig. 9. The correlations between the simulated mean yields by different models and environmental variables, which are used to create the strata for the StrRS. The terrain is the elevation of the grid cell referring to the sea level. Temperature means the daily mean temperature. Precipitation and solar radiation are the mean annual sum of them. SWHC down to -1.5 indicates the soil water holding capacity down to -1.5 m.

into strata may improve the sampling precision, which was not considered in the present study (Brus, 1994; Cochran, 1977). For example, Hirzel and Guisan (2002) compared the equal-stratified with the proportional-stratified random sampling scheme in modeling the habitat suitability and found the equal-stratified scheme could achieve a higher precision. This conclusion needs to be further validated in modeling crop yield at a regional scale. In addition, for the compact geographical stratification, the *k-mean++* algorithm cannot guarantee that the cluster will be equal size in the case that the population size is divisible by the sample size, even that we ran the algorithm with 100 different centroid seeds. To overcome this problem, a R package named *spcosa* is implemented which can enforce clusters (strata) of equal size (compact geographical strata of equal size) (Walvoort et al., 2010). Using this specially designed package may improve the performance of compact geographical stratification. Furthermore, this study only considered the design-based sampling strategies (e.g. SimRS and StrRS) (Wang et al., 2002). The model-based sampling strategies such as the Kriging method have been frequently applied to optimize the spatial sampling (Brus and Heuvelink, 2007; Theodossiou and Latinopoulos, 2006). A comparison of these two classes of sampling methods can further improve the selection of simulation sites for crop modeling over a large area. Finally, we compared the precision of different sampling schemes in estimating the mean yields over a region, while the spatial structure and inter annual variability of the yields are among other aspects that can be of potential interests for some applications (Fortin et al., 1989).

5. Conclusions

This study investigated how the simulation sites should be distributed among the varying environmental conditions over a large study area, when estimating the mean of simulated crop yields of a region. We found that stratified random sampling (StrRS) had a large positive precision gain over the simple random sampling (SimRS) only when the strata were created based on one of the most influential environmental variables of the respective crop model. Using the compact geographical stratification can modestly improve the sampling precision regardless of crop models and sample sizes. The efficiency of StrRS can be improved by large sample size, large number of strata and by stratification of the population or study area based on the most influential environmental variable.

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Appendix A. Supplementary data

Supplementary data related to this article can be found at <http://dx.doi.org/10.1016/j.envsoft.2016.02.022>.

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